

1 Solution — GIAM 1.4

Problem 9

1.1 Problem

It is a well-known fact that if a number is divisible by **3**, then **3** divides the sum of its (decimal) digits. Is this result true in base **7**? Do you think this result is true in *any* base?

1.2 Short Answers

- **In base 7: yes.** The digit-sum test for **3** works in base **7** exactly as it does in decimal.
- **In *any* base: no.** It holds precisely when the base b satisfies $b \equiv 1 \pmod{3}$ — true for 10 and for 7, but false for many other bases.

1.3 Why the Decimal Rule Works

The rule is really a fact about **10** modulo **3**, not about decimal as such. Write a number in base b :

$$n = d_k b^k + \cdots + d_1 b + d_0.$$

Reduce modulo **3**. If $b \equiv 1 \pmod{3}$, then $b^i \equiv 1^i = 1 \pmod{3}$ for every i , so each place value contributes just its digit:

$$n \equiv d_k + \cdots + d_1 + d_0 = \sum_i d_i \pmod{3}.$$

Hence $3 \mid n$ if and only if $3 \mid \sum_i d_i$. For decimal, $10 \equiv 1 \pmod{3}$, which is exactly the hypothesis — so the familiar rule follows.

1.4 Base 7: Yes

Everything above used only the single fact $b \equiv 1 \pmod{3}$. For base **7**:

$$7 = 2 \cdot 3 + 1 \equiv 1 \pmod{3},$$

so $7^i \equiv 1 \pmod{3}$ for all i , and the *same* argument gives $n \equiv \sum_i d_i \pmod{3}$. Therefore in base 7, too, **a number is divisible by 3 exactly when its base-7 digit sum is.**

1.5 Example

Take $3267 = 12345_7$ (from Problem 4). Its base-7 digit sum is

$$1 + 2 + 3 + 4 + 5 = 15, \quad 3 \mid 15,$$

so the rule predicts $3 \mid 3267$ — and indeed $3267 = 3 \cdot 1089$. (The decimal digit sum $3 + 2 + 6 + 7 = 18$ is also divisible by 3, agreeing.)

1.6 Any Base? No — It Depends on $b \bmod 3$

The test is *not* universal. The behaviour splits into three cases according to $b \bmod 3$:

The digit-sum test for 3 works exactly when every power of the base is $\equiv 1 \pmod{3}$.

Why decimal and base 7 both work:

	b^0	b^1	b^2	b^3	$\pmod{3}$
base 10:	1	10	100	1000	1, 1, 1, 1
base 7:	1	7	49	343	1, 1, 1, 1

Both have base $\equiv 1 \pmod{3}$, so $n = \sum d_i \cdot b^i \equiv \sum d_i \pmod{3} \rightarrow$ the digit-sum test holds.

But not in every base — it depends on $b \bmod 3$:

base $b \bmod 3$	test for divisibility by 3	example bases
$\equiv 1$	digit sum (the familiar rule)	4, 7, 10, 13, 16
$\equiv 2$	alternating digit sum	2, 5, 8, 11
$\equiv 0$	last digit (since $3 \mid b$)	3, 6, 9, 12

Example: $3267 = 12345_7$. Digit sum $1+2+3+4+5 = 15$, and $3 \mid 15 \rightarrow 3 \mid 3267$. ✓

Figure 1.1: Top: powers of 10 and of 7 are all $\equiv 1 \pmod{3}$, so $n \equiv$ digit sum $\pmod{3}$ in both bases — the digit-sum test holds. Bottom: classified by $b \bmod 3$, divisibility-by-3 uses the digit sum ($b \equiv 1$: bases 4, 7, 10, ...), the alternating digit sum ($b \equiv 2$: bases 2, 5, 8, ...), or the last digit ($b \equiv 0$: bases 3, 6, 9, ...). Example: $3267 = 12345_7$, digit sum 15.

- $b \equiv 1 \pmod{3}$ (bases 4, 7, 10, 13, ...): $b^i \equiv 1$, so the **plain digit sum** works — this is the decimal/base-7 case.
- $b \equiv 2 \equiv -1 \pmod{3}$ (bases 2, 5, 8, 11, ...): $b^i \equiv (-1)^i$, so $n \equiv \sum_i (-1)^i d_i \pmod{3}$ — an **alternating** digit sum, not the plain one.
- $b \equiv 0 \pmod{3}$ (bases 3, 6, 9, ...): here $3 \mid b$, so $b^i \equiv 0$ for $i \geq 1$ and $n \equiv d_0 \pmod{3}$ — a **last-digit** test.

So the digit-sum rule for 3 holds **iff** $b \equiv 1 \pmod{3}$. It is a happy accident that both of our everyday-ish bases, 10 and 7, fall into that class.

1.7 Remark — The Same Principle Behind Every Divisibility Rule

This is the general fact already met in Problem 5 (octal evenness), now applied to $m = 3$. For a modulus m and base b :

relationship of b to m	reduction of b^i	resulting test
$m \mid b$	0	last digit ($n \equiv d_0$)
$b \equiv 1 \pmod{m}$	1	digit sum
$b \equiv -1 \pmod{m}$	$(-1)^i$	alternating digit sum

Table 1.1

The “well-known” decimal digit-sum rule for 3 is just the middle row with $m = 3$, $b = 10$. Reading the table the other way explains *all* the standard tricks at once: divisibility by 2 or 5 in decimal is a last-digit test (both divide 10); divisibility by 11 is an alternating-sum test ($10 \equiv -1 \pmod{11}$).

1.8 Remark — The Rule for 9 Comes Free

Because $10 \equiv 1 \pmod{9}$ as well, the *identical* argument gives the digit-sum test for 9 in decimal: $9 \mid n \iff 9 \mid \sum_i d_i$. This is the basis of the old arithmetic check “**casting out nines.**” It transfers to base 7 only if $7 \equiv 1 \pmod{9}$ — which is false ($7 \not\equiv 1$) — so the digit-sum-for-9 rule does *not* hold in base 7. (What *does* hold in base 7 is a digit-

sum test for 6, since $7 \equiv 1 \pmod{6}$.) The moral: each base inherits digit-sum tests for exactly the divisors m of $b - 1$.

1.9 Remark — Divisibility by 3 in Binary

Binary is the most familiar base in the *alternating-sum* class, since $2 \equiv -1 \pmod{3}$. So a binary number is divisible by 3 iff its **alternating bit sum** is:

$$6 = 110_2 : 0 - 1 + 1 = 0 \pmod{3}, \quad 9 = 1001_2 : 1 - 0 + 0 - 1 = 0 \pmod{3}, \quad 5 = 101_2 : 1 - 0 + 1 = 2 \pmod{3}$$

The first two are multiples of 3, the third is not — correctly. So even where the *plain* digit-sum test fails, a close cousin (the alternating sum) takes over; the underlying machinery, $b^i \pmod{m}$, never stops working.

1.10 Remark — It Was Never About Decimal

The cleanest takeaway is that the “well-known fact” is mis-stated when pinned to decimal. The real theorem is:

*In base b , divisibility by m can be checked from the digit sum **whenever** $b \equiv 1 \pmod{m}$.*

Decimal earns the digit-sum tests for 3 and 9 only because $10 - 1 = 9$ is divisible by 3 and 9. Base 7 earns them for 3 and 6 (and 2) because $7 - 1 = 6$. The rule “feels” like a property of decimal arithmetic, but it is really a property of the number $b - 1$ — which is why the question “is it true in base 7?” has the satisfying answer *yes, for the very same reason*, and “in any base?” the answer *no, only when $b \equiv 1 \pmod{3}$* .